

PAYMENT ENGINE - API v1

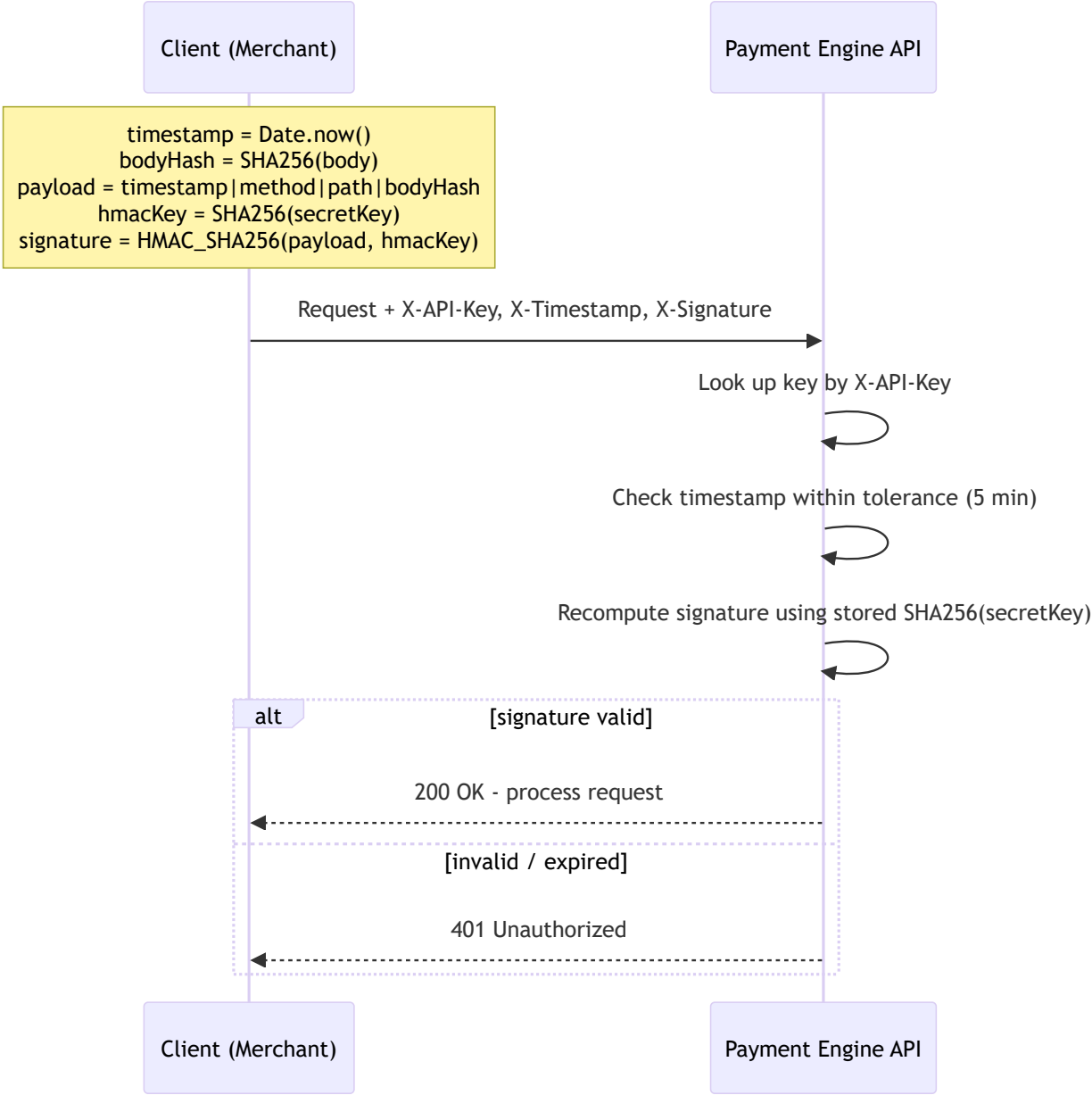
Payment Flow Charts

Crypto-to-fiat payment processing: authentication, status lifecycle,
and payment type flows (transfer, gift, request, merchant)

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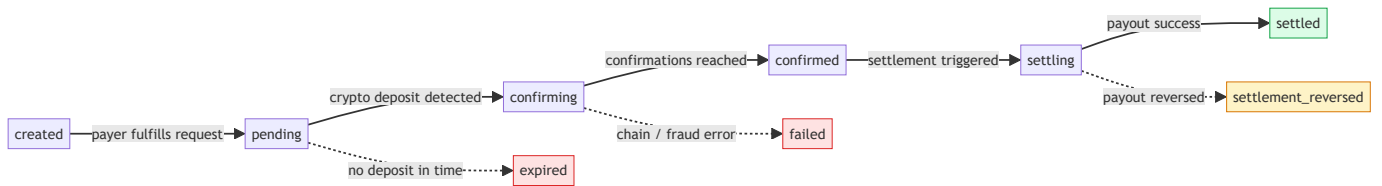
1. HMAC Request Authentication

Every call to a protected `/v1/payments*` endpoint is signed with HMAC-SHA256. The signing key is `SHA256(secretKey)`, never the raw secret, so the server only ever stores a hash.



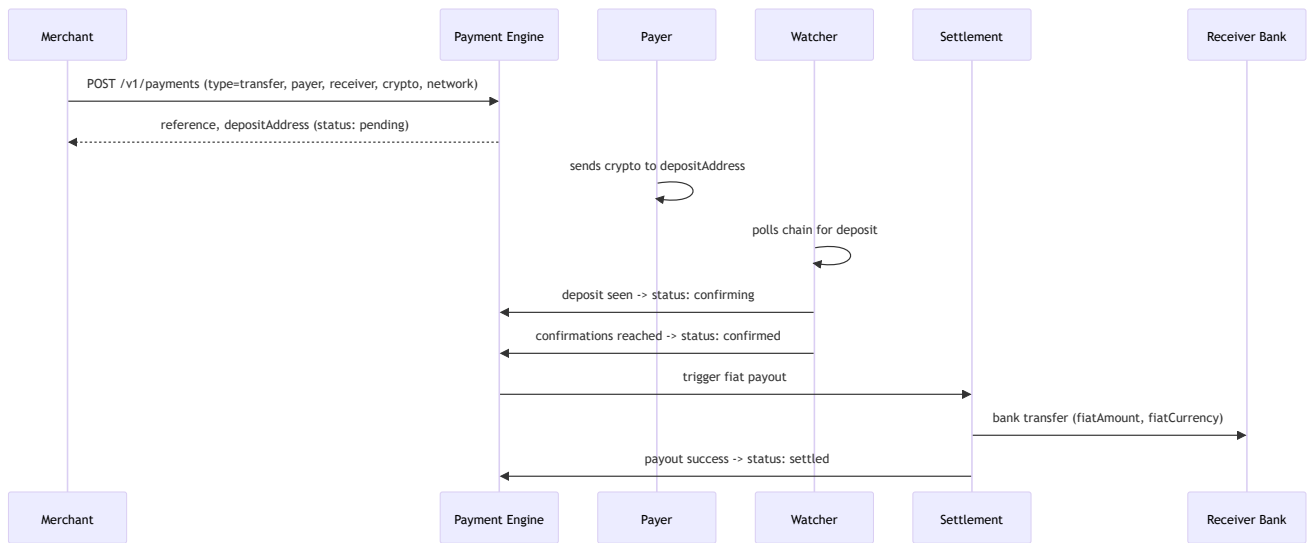
2. Payment Status Lifecycle

All payment types converge on this state machine. Transfers and gifts start at `pending` (deposit address issued immediately). Requests without pre-set crypto start at `created` and move to `pending` once fulfilled.



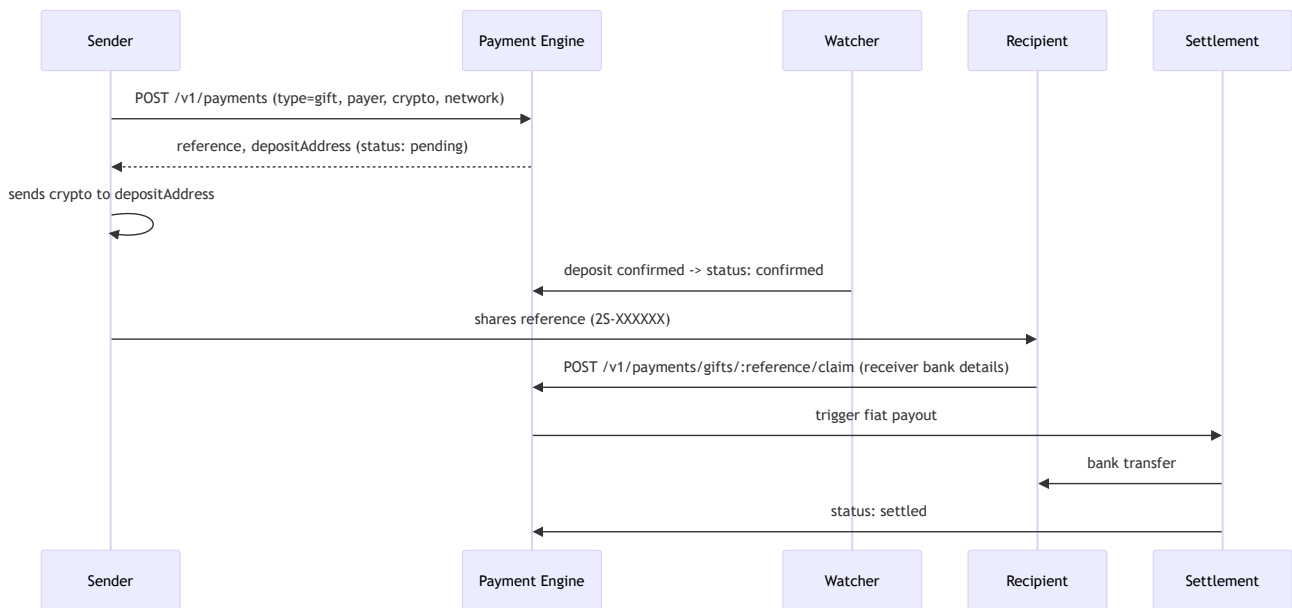
3. Transfer Flow

Direct payment: both payer and receiver are known up front. `POST /v1/payments` with `type: "transfer"`.



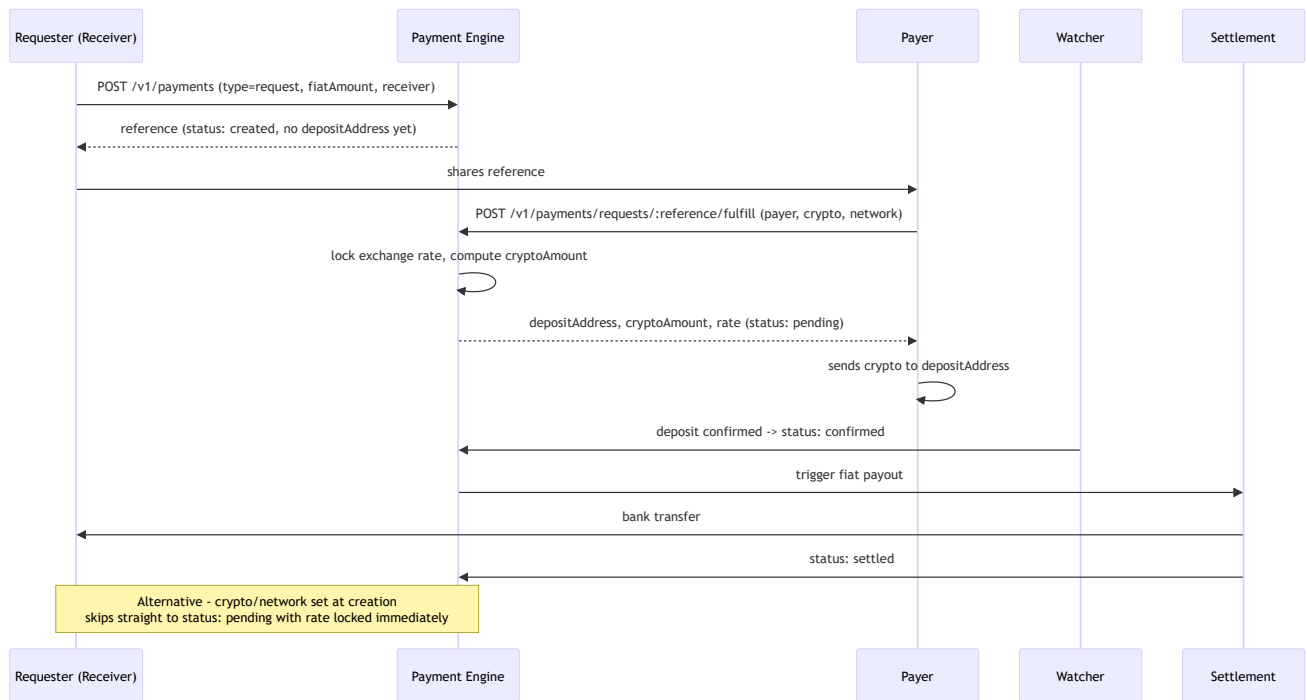
4. Gift Flow

Sender pays crypto without a receiver yet; a recipient claims the gift later using the `reference` and supplies their own bank details.



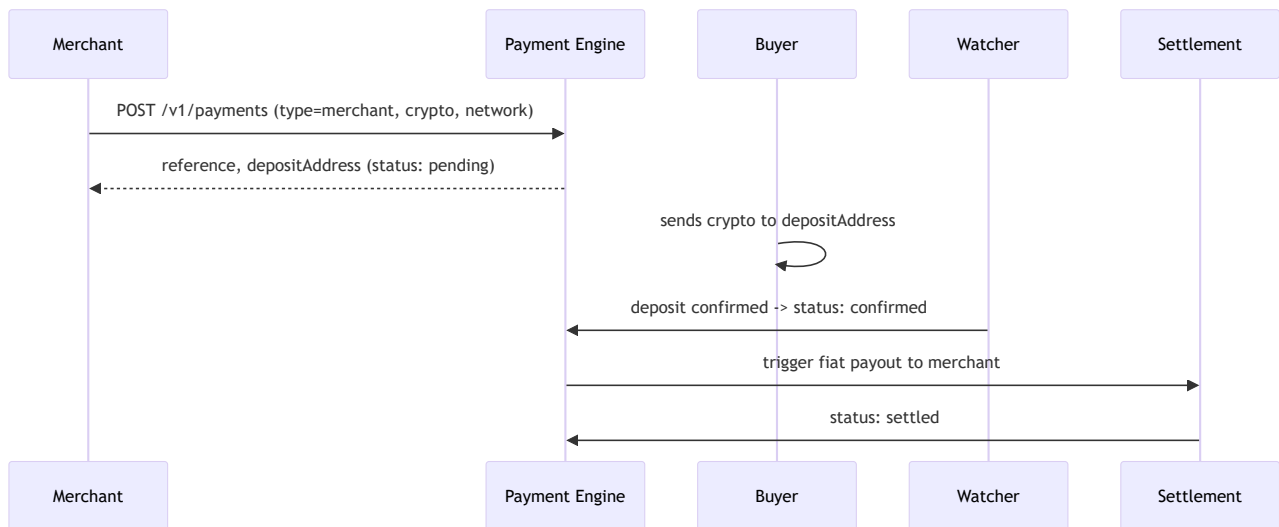
5. Request Flow

Receiver specifies only a fiat amount; the payer chooses crypto/network at fulfillment time and the exchange rate is locked then. Crypto can optionally be pre-set at creation, which skips straight to **pending**.



6. Merchant Flow

Merchant checkout: crypto/network required, payer and receiver both optional (used for storefront-style checkouts settling to the merchant's own configured payout account). Structurally follows the transfer path once a deposit address is issued.



7. Payment Type Comparison

Type	Payer	Receiver	Crypto/Network	Initial Status	Use Case
transfer	Required	Required	Required	pending	Direct payment
gift	Required	Optional (claim later)	Required	pending	Send crypto gift
request	Optional (fulfill later)	Required	Optional	created or pending	Payment invoice
merchant	Optional	Optional	Required	pending	Merchant checkout